Total power p = 3 × power in each phase
Power in each phase $P = V_{ph}\ I_{ph}\ \cos\theta$

$$P = 3 \times \frac{V_L}{\sqrt{3}} \times I_L \cos\theta$$

$$\text{(where as } I_{ph} = I_L,\ V_{ph} = \frac{V_L}{\sqrt{3}}\text{)}$$

$$\therefore P = \sqrt{3} V_L I_L \cos\theta$$

5.5.6. Delta or Mesh Connection(Δ)

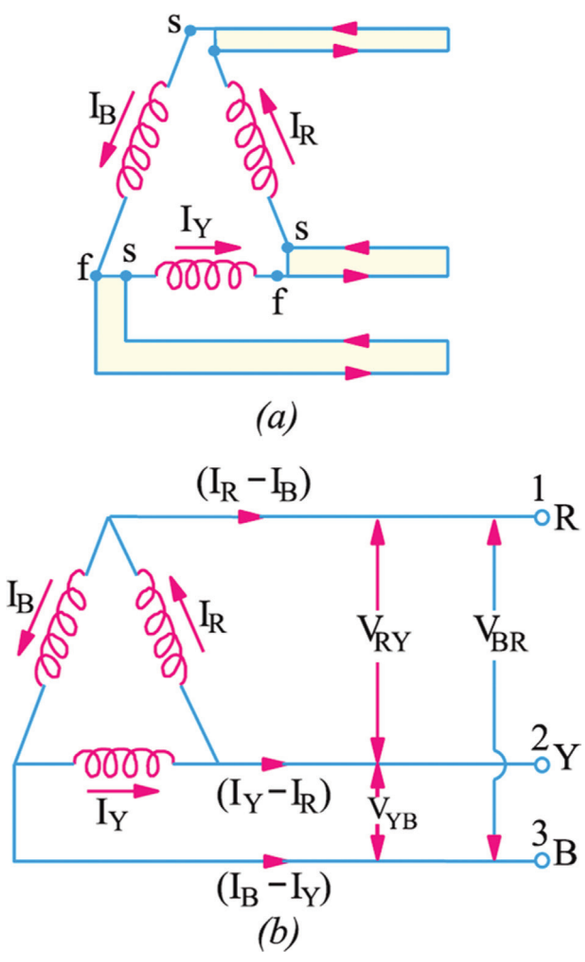


Fig 5.15 Delta Connection

In this method of interconnection, the dissimilar ends of the three phase windings are joined together. The finishing end of one phase is connected to the starting end of the other phase so as to obtain Mesh or Delta connection. The three line conductors are taken from the three junctions of the Mesh or Delta and they designated as R, Y and B. This is called three phase three wire delta connected system. Since no neutral exists in a Delta connection, only three phase, three wire system can be formed.

In this connection, the line voltage is equal to the phase voltage.

Phase Volltage = Line voltage

$$V_{ph} = V_L$$

Phase Current = Line current

$$I_{ph} = \frac{I_L}{\sqrt{3}}$$

POWER

Total power P = 3 × power per phase

$$P = 3\ V_{ph}\ I_{ph} \cos\theta$$

$$P = 3 \times V_L \times \frac{I_L}{\sqrt{3}} \times \cos\theta$$

$$V_{ph} = I_L,\ I_{ph} = \frac{I_L}{\sqrt{3}}$$

$$\text{i.e., Power} = \sqrt{3}\ V_L I_L \cos\theta$$

Where cosθ is power factor.

5.5.7 Advantages of Star Connection Over Delta Connected System

A star connected alternator will require less number of turns than a delta connected alternator for the same line voltage.

A star connected alternator requires less insulation over a Delta connected alternator for the same line voltage.

In star connection, Three-phase, Four-wire system permits to use two voltage ie, Phase voltage as well as line voltage.

In star connection single phase loads can be connected between any one

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